Electromagnetism and Special Relativity

Directed Reading Program in Mathematics

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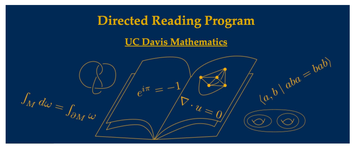


In 1860, physicist James Maxwell deciphered a combination of equations indicating that light propagates at a constant speed, based on universal constants of electromagnetism. With this idea in mind, and the support of numerous experiments leading to the conclusion that the speed of light was constant, Albert Einstein developed a theory we now call special relativity. Special relativity ensures the speed of light remains constant for all observers regardless of their speed. This challenged the long held belief that speed is relative and time is absolute, and demonstrated that the version of relativity presented by Galileo was incomplete.

Electromagnetism: Speed of light Value

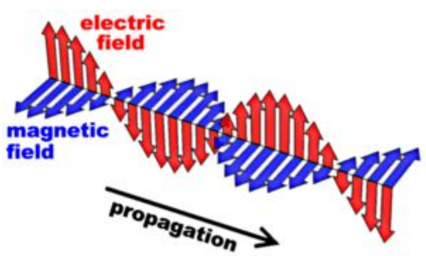
We examine Gauss' Law, Faraday's Law, and Ampere's Law which are collectively known as Maxwell's equations. In the absence of charges and therefore currents, these equations imply that the electric and magnetic field satisfy a famous partial differential equation.

$$\nabla \cdot \vec{\mathbf{E}} = 0 \qquad \nabla \cdot \vec{\mathbf{B}} = 0$$

$$\nabla \times \vec{\mathbf{E}} = -\frac{\partial \vec{\mathbf{B}}}{\partial t} \qquad \nabla \times \vec{\mathbf{B}} = \mu_0 \varepsilon_0 \frac{\partial \vec{\mathbf{E}}}{\partial t}$$

$$\nabla \times (\nabla \times \vec{\mathbf{A}}) = \nabla(\nabla \cdot \vec{\mathbf{A}}) - \nabla^2 \vec{\mathbf{A}}$$

$$\nabla^2 \vec{\mathbf{E}} = \mu_0 \varepsilon_0 \frac{\partial^2 \vec{\mathbf{E}}}{\partial t^2} \qquad \nabla^2 \vec{\mathbf{B}} = \mu_0 \varepsilon_0 \frac{\partial^2 \vec{\mathbf{B}}}{\partial t^2}$$



Wave equation form

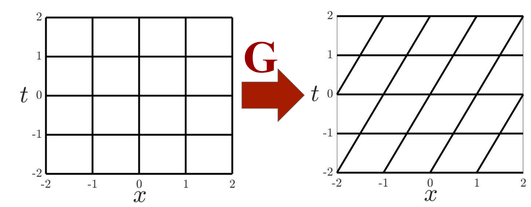
$$\nabla^2 f = \frac{1}{v^2}\frac{\partial^2 f}{\partial t^2} \qquad c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$$

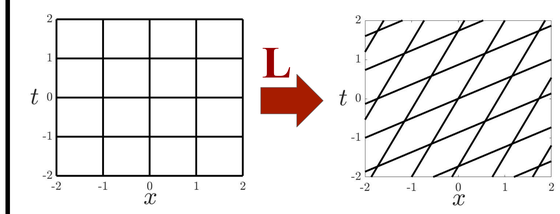
Lorentz Transformations: Conservation of "c"

Galilean Transformations — When velocities are small compared to the speed of light — **Lorentz Transformations**

$$\begin{bmatrix} t' \\ x' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ v & 1 \end{bmatrix} \begin{bmatrix} t \\ x \end{bmatrix}$$

$$\beta = \frac{v}{c} \qquad \gamma = \frac{1}{\sqrt{1-\beta^2}} \qquad \begin{bmatrix} ct' \\ x' \end{bmatrix} = \begin{bmatrix} \gamma & \beta\gamma \\ \beta\gamma & \gamma \end{bmatrix} \begin{bmatrix} ct \\ x \end{bmatrix}$$





$$\frac{dx'}{dt'} = v + \frac{dx}{dt}$$

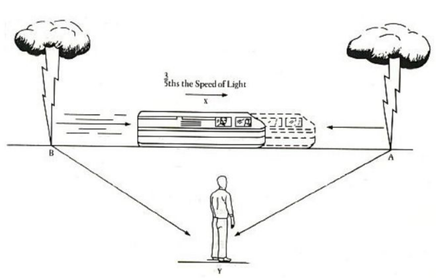
- Velocities are all shifted by v.
- Simultaneity is preserved.

$$\frac{dx'}{dt'} = \frac{v + \frac{dx}{dt}}{1 + \frac{\beta}{c}\frac{dx}{dt}}$$

- Speed of light is invariant.
- Simultaneity is not preserved.

Relativity, Invariance, and Covariance Concepts

By understanding that the speed of light is governed by universal constants related to electromagnetism, it is also necessary to treat this value as a constant, and that having no mass would mean that nothing could go faster than light. This idea will allow for the exploration of new relativistic concepts.



- ***Covariance:*** ***Equations that do not change regardless of the frame of reference, such as the "conservation of momentum".***
- ***Invariance:*** ***Quantities that do not change regardless of the frame of reference, such as mass or charge.***

Thought Experiment:

Suppose a person is on a train that is traveling at a speed relatively close to the speed of light. Two thunderclaps occur, one in front of the train and one behind it.

Assuming the speed of light is constant, one can say that the person on the train will see the flash of light from the front thunderclap occurring before the flash of light from the thunderclap behind.

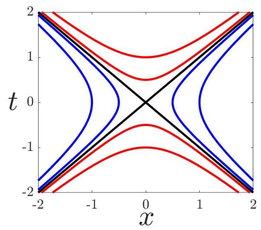
However, if we look at the person in the middle, they witnesses both thunderclaps occurring simultaneously.

How is it possible that one person sees two events happening at the same time while another sees them happening at different times, if both are living in the same reality?

New Geometry of a Relativistic "Dot Product"

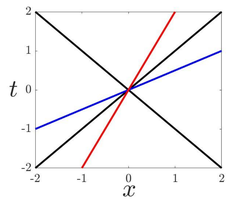
$$(t_1, x_1)\cdot(t_2, x_2) = -t_1 t_2 + x_1 x_2 = \begin{bmatrix} t_1 & x_1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} t_2 \\ x_2 \end{bmatrix}$$

Lorentz transformations have another invariant, the "spacetime interval". This looks like a traditional dot product, but the time variables get multiplied by negative one.



Trichotomy:

- **Space-like:** Dot product value is larger than 0.
- **Time-like:** Dot product value is less than 0.
- **Light-like**: Dot product value is equal to 0.



With this new concept of the relativistic dot product, the definition of orthogonality is now based on how a line belonging to the space-like lines space is reflected in the time-like lines space and vice versa with respect to light-like lines.

Connections with Calculus

$\mathbf{r} : \mathbb{R} \to \mathbb{R}^2$ a regular spacetime curve means $\mathbf{r}' \cdot \mathbf{r}' \neq 0$

Like in 21D parametrize so that $\mathbf{r}' \equiv \mathbf{T}$ has $\mathbf{T} \cdot \mathbf{T} = \pm 1$.

Then $\mathbf{T}' = \kappa(s)\mathbf{N}$ and $\mathbf{T} \cdot \mathbf{T} = \pm 1 \iff \mathbf{N} \cdot \mathbf{N} = \mp 1$.

Proposition: Relativistic Frenet formula $\begin{bmatrix} \mathbf{T}' \\ \mathbf{N}' \end{bmatrix} = \begin{bmatrix} 0 & \kappa \\ \kappa & 0 \end{bmatrix} \begin{bmatrix} \mathbf{T} \\ \mathbf{N} \end{bmatrix}$

Proof: $\mathbf{T} \cdot \mathbf{N} = 0$ means $0 = (\mathbf{T} \cdot \mathbf{N})' = \mathbf{T}' \cdot \mathbf{N} + \mathbf{T} \cdot \mathbf{N}'$
$= \kappa \mathbf{N} \cdot \mathbf{N} + \mathbf{T} \cdot \mathbf{N}'$

Therefore $\mathbf{T} \cdot \mathbf{N}' = -\kappa \frac{\mathbf{N} \cdot \mathbf{N}}{\mathbf{T} \cdot \mathbf{T}} = \kappa$ which means $\mathbf{N}' = \kappa \mathbf{T}$

$$\nabla \cdot \vec{\mathbf{E}} = \frac{\rho}{\varepsilon_0}$$

$$\nabla \cdot \vec{\mathbf{E}} = 0$$

$$\nabla \cdot \vec{\mathbf{B}} = 0$$

$$\nabla \times \vec{\mathbf{B}} = \mu_0 \varepsilon_0 \frac{\partial \vec{\mathbf{E}}}{\partial t}$$

$$\nabla \times \vec{\mathbf{B}} = \mu_0 \vec{\mathbf{J}} + \mu_0 \varepsilon_0 \frac{\partial \vec{\mathbf{E}}}{\partial t}$$

$$\nabla \times \vec{\mathbf{E}} = -\frac{\partial \vec{\mathbf{B}}}{\partial t}$$

$$\nabla^2 f = \frac{1}{v^2} \frac{\partial^2 f}{\partial t^2}$$

$$\nabla \times (\nabla \times \vec{\mathbf{A}}) = \nabla(\nabla \cdot \vec{\mathbf{A}}) - \nabla^2 \vec{\mathbf{A}}$$

$$\nabla^2 \vec{\mathbf{E}} = \mu_0 \varepsilon_0 \frac{\partial^2 \vec{\mathbf{E}}}{\partial t^2} \qquad \nabla^2 \vec{\mathbf{B}} = \mu_0 \varepsilon_0 \frac{\partial^2 \vec{\mathbf{B}}}{\partial t^2}$$

$$\zeta = \tanh^{-1}\beta$$

$$L = \begin{bmatrix} \cosh\zeta & \sinh\zeta \\ \sinh\zeta & \cosh\zeta \end{bmatrix}$$

$$\gamma = \frac{1}{\sqrt{1-\beta^2}} \qquad \beta = \frac{v}{c}$$

$$\begin{bmatrix} ct' \\ x' \end{bmatrix} = \begin{bmatrix} \gamma & \beta\gamma \\ \beta\gamma & \gamma \end{bmatrix} \begin{bmatrix} ct \\ x \end{bmatrix}$$

$$\mathbf{r} : \mathbb{R} \to \mathbb{R}^2$$
$$\mathbf{r}' \cdot \mathbf{r}' \neq 0$$

$$\vec{\mathbf{T}}(s) = \vec{\alpha}'(s) \qquad \vec{\mathbf{T}}'(s) = \kappa(s)\vec{\mathbf{N}}(s)$$

$$\vec{\mathbf{T}} \cdot \vec{\mathbf{T}} = \pm 1 \implies \vec{\mathbf{N}} \cdot \vec{\mathbf{N}} = \mp 1$$

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